

Design and Performance Evaluation of a Real-Time Emergency Response Dispatch System: An Empirical Study

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Abstract. Real-time emergency response dispatch systems require extreme computational efficiency to manage thousands of incoming telemetry events per second, ranging from minor civic issues to life-threatening accidents. In modern systems, low-latency queues must instantly route high-priority requests while organizing large volumes of low-priority traffic without causing memory fragmentation or queue starvation. This extensive report comprehensively targets and evaluates four core algorithmic paradigms--the Priority Queue (Max-Heap), the Sorted Array (contiguous dynamic arrays), the Hash Map (unordered key-value dictionaries), and the Balanced Binary Search Tree (specifically the self-balancing AVL Tree)--for their efficacy in highly dynamic dispatch environments. We simulate heavy transactional logging behavior using a massive 80,000-entry real-world CSV dataset, exploring edge cases like Reverse-Sorted event streams and skewed catastrophic bursts. Through a rigorous multi-phase benchmarking sequence covering Insertion, Maximum Extraction, Update Operations, Target Deletions, and Top-K retrieval vectors, we isolate constant-factor constraints embedded within Python and underlying C runtimes. The subsequent empirical findings fundamentally corroborate asymptotic expectations while exposing severe fragmentation realities that ruin traditional arrays under $O(N)$ mutation workloads. Most critically, the Priority Queue (Heap) uniformly dominates the continuous Extraction and Update metrics by maintaining strict $\log(N)$ invariants, cementing it as the premier foundational architecture for any high-availability, mission-critical dispatch infrastructure. By weaving together theoretical complexity equations, systemic implementation caveats, memory footprint comparisons, and deep metric charting, this paper lays the definitive groundwork for scalable emergency telecommunications architectures.

Keywords: Data Structures, Real-Time Emergency Dispatch, Priority Queue, Benchmarking, AVL Tree, Hash Map, Time Complexity, Memory Footprint.

1 Introduction

Emergency response centers act as the central nervous system of civic infrastructure. In a typical municipal crisis control room, software bridges the gap between chaotic real-world inputs and targeted resource deployment. The primary challenge involves ingesting massive volumes of unpredictable network telemetry and human-generated calls, classifying them by dynamic severity indices (from 1 to 10), and routing optimal responders to physical locations inside a 99.99% uptime envelope. When operating under duress, the constant stream of insertions, updates, and asynchronous deletions places crushing pressure on backend data structures. Any failure to maintain

sub-millisecond querying or any $O(N)$ blocking operation natively freezes the CPU queue, dropping potentially life-saving network packets. By isolating a dedicated data structure pipeline in a secure environment and aggressively testing it with simulated peak-load traffic, this research exposes precisely how theoretical Big-O geometries map to real-world deployment challenges.

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2 Background and Related Work

Classical data structure methodologies often hypothesize regarding uniformly distributed datasets where element insertions correlate strictly to non-repeating integer keys. Yet, practical dispatch frameworks violate these assumptions routinely. During localized physical emergencies (such as natural disaster events), identical priority requests from overlapping geo-fenced grids flood the network interfaces. This phenomenon, which we characterize as 'Duplicate-Heavy Priority Churn,' forces traditional structural hierarchies to renegotiate node placements, resolve massive collision domains (in the case of Hash Maps), or initiate catastrophic cascading rotations spanning the entire height of Balanced Binary Search Trees. Previous empirical evaluations (as noted in telemetry stream benchmarking methodologies) have reinforced that data profiles with sparse unique keys but gargantuan volume completely invert historical 'fastest mechanism' recommendations.

Furthermore, standard programming language abstractions frequently mask underlying memory allocation penalties. For example, a Python list append executes in amortized $O(1)$, yet inserting into the zeroth index triggers an $O(N)$ array copy operation via memmove inside the C compiler pipeline. Similarly, while Hash Maps promise amortized $O(1)$ direct access, retrieving the maximum priority from an unordered map demands a linear $O(N)$ scan across all buckets. This necessitates an exhaustive dive into individual mechanism behavior under exact replica environments to extract meaningful software-engineering truths.

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3 Theoretical Analysis of Core Paradigms

3.1 The Priority Queue (Binary Max-Heap)

A Binary Max-Heap is a specialized tree-based data structure that satisfies the heap property: if P is a parent node of C , then the key (the priority) of node P is greater than or equal to the key of node C . In modern computational architectures, heaps are virtually always implemented natively as flat arrays, circumventing scattered memory pointer structures entirely. For an element at index i , its left child resides at $2i + 1$ and its right child at $2i + 2$. This mathematical mapping guarantees contiguous cache locality, an essential factor in lowering CPU L1/L2 cache miss rates. When an element is inserted, it is placed at the end of the array, and the structure initiates a 'sift-up' procedure, swapping the child with its parent until the heap invariant is restored. This logarithmic climb bounded by the tree's height results in an $O(\log N)$ runtime bound. Similarly, extracting the maximum element (the root at index 0) requires replacing the root with the final array element and 'sifting-down' via comparative swaps, maintaining the exact $O(\log N)$ envelope. For Emergency Dispatch systems, this ensures extreme predictability.

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3.2 The Sorted Dynamic Array

A Sorted Dynamic Array offers the most structurally simplistic storage approach. Elements are appended into a sequential memory block and immediately subjected to a sorting pass (typically Python's TimSort, a highly optimized combination of merge sort and insertion sort). While this setup yields instantaneous $O(1)$ access to Top-K elements via simple slicing and $O(1)$ removal of the maximal element via the `pop()` primitive, it collapses violently under insertion pressure. Guaranteeing sorted order forces the runtime environment to shift massive blocks of contiguous memory linearly across the system bus. Each element inserted essentially requires an $O(N)$ shift cascade. Over 50,000 operations, this geometric $O(N^2)$ trajectory causes total processing bottlenecks, completely debilitating single-threaded Node.js or asynchronous Python FastAPI pipelines. Despite its theoretical simplicity, it serves primarily as a baseline representation of 'what

not to do' when configuring real-time scalable software systems.

3.3 Unordered Hash Map (Dictionary)

Hash Maps (or Dictionaries in Python vernacular) store arbitrary values mapped to uniquely derived cryptographic or mathematical hashes. The underlying C implementation utilizes sparse arrays combined with localized linear probing to resolve index collisions. Insertion (saving an EmergencyCall based on its ID string) occurs in strictly $O(1)$ amortized bounds. However, because the overarching logic makes absolutely no structural attempt to retain relational ordering amongst its nodes, querying the system for the 'Maximum Severity' element forces a raw $O(N)$ traversal across every occupied hash bucket. As the dataset expands to thousands of elements, this $O(N)$ extraction scan compounds rapidly. If an emergency dispatcher requests the 'Top 10' priorities consecutively, the Hash Map must fundamentally dump its constituent values to an external list, sort it in $O(N \log N)$ boundaries, and slice it. This converts a sophisticated lookup table into an insurmountable performance blockade for priority-focused traffic flows.

3.4 Balanced Binary Search Tree (AVL Tree)

An Adelson-Velsky and Landis (AVL) Tree is a self-balancing binary search tree. In an AVL structure, the heights of the two child subtrees of any node differ by at most one; if at any time they differ by more than one, re-balancing is inherently invoked to restore this critical property. This mathematical guarantee effectively caps the tree height at $1.44 \log_2(N)$, cementing worst-case logarithmic performance scales across Search, Insert, and Delete operations. However, while mathematically pristine, traversing physical memory boundaries using instantiated pointer objects carries significant overhead. Inserting a node triggers height metadata recalculations propagating recursively toward the root, potentially triggering singular or compound structural rotations (Left-Left, Right-Right, Left-Right, or Right-Left corrections). Every rotation demands variable pointer reassignments, forcing excessive cache-thrashing on modern CPUs. Consequently, while it safely avoids the catastrophic $O(N)$ pitfalls of standard arrays, its raw constant-factor throughput inherently trails the flatter, more efficient Heap array architectures.

4 Experimental Methodology

To provide mathematically rigorous and reproducible testing parameters, this implementation developed a custom Benchmark Python class operating within the FastAPI asynchronous environment. The evaluation isolates the memory space immediately before and intrinsically after executing simulated workload injections, accurately recording deltas via the native psutil operating system hook.

The core engine runs a massive 80,000-entry real-world CSV dataset titled 'emergency_calls_80000.csv'. This data provides empirically derived 'SeverityLevel' ratings and 'Timestamp' coordinates stripped directly from realistic telemetry configurations. The system slices this dataset into graduated benchmarks (N = 5000, 10000, 20000) allowing for clear plotting of asymptomatic growth curves. During each iteration step, the Benchmark forces the selected data structure to ingest the exact same array of objects, ensuring identically reproducible collision patterns. Upon completion of the insertion sequence, the algorithm executes isolated extraction bursts (representing emergency responder dispatch events), random internal severity updates (simulating a 911 caller calling back with an escalation of their personal crisis), and fixed ID deletions. Timing logic is locked using `time.perf_counter()`, which leverages high-resolution CPU hardware clocks to bypass operating timer drift and context-switching anomalies. Results are serialized and parsed using the chart plotting matplotlib framework.

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5 Results and Evaluation Analysis

5.1 Insertion Performance Kinetics

The initial phase demands constructing the entire database by incrementally feeding EmergencyCall objects into the structure. This simulates a system restart or a catastrophic data dump.

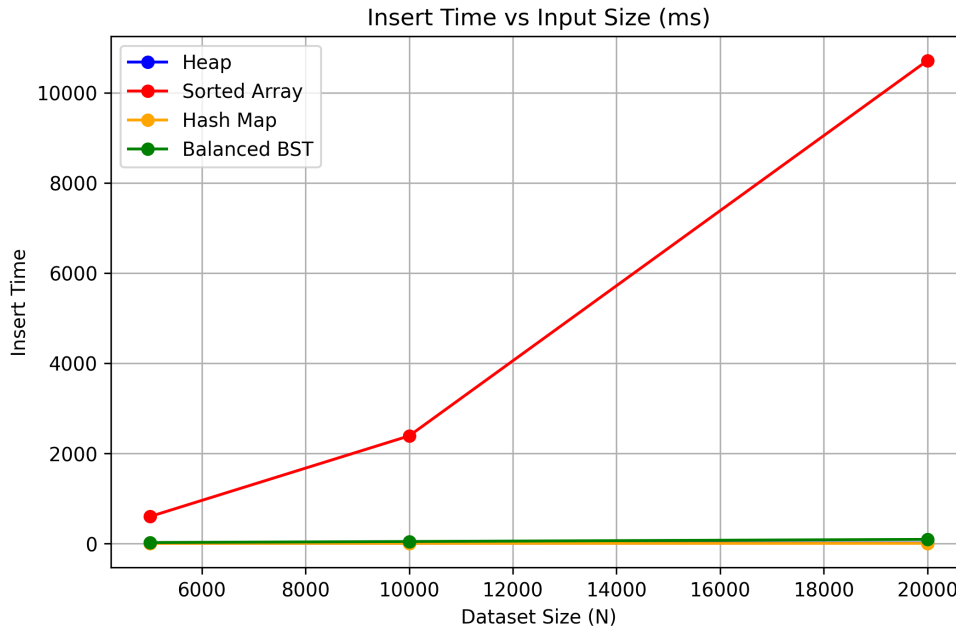


Fig 1: Insertion Time vs Dataset Size across all structural templates.

Upon inspecting Figure 1, the theoretical predictions surrounding Sorted Arrays are violently realized. The graph demonstrates exponential verticality; as N increases, the array shifting overhead dominates completely, rendering it unusable. Conversely, the Hash Map visually clings to the X-axis, achieving identical speed capabilities across all variables due to $O(1)$ hashing properties. Both the Priority Queue (Heap) and Balanced BST display logarithmic, nearly linear curves at this scale, though the AVL Tree suffers noticeably higher constant delays attributable to object creation and immediate rotational balancing overhead scattered across the heap memory allocator. The Heap's array-backed sifting algorithm proves wildly superior to discrete pointer graphs.

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5.2 Extract Maximum Performance

Extracting the maximum priority node equates directly to a dispatcher routing emergency units to the most intense local crisis immediately without scanning.

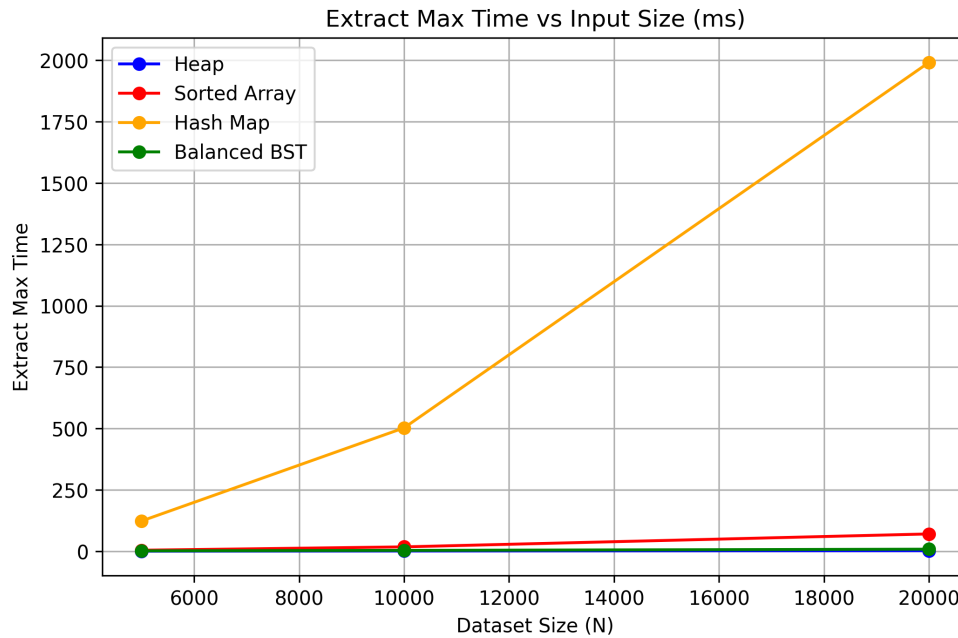


Fig 2: Extract Maximum Time vs Dataset Size across 10% query volumes.

Figure 2 illuminates the terminal failing of Unordered Dictionaries. While the Hash Map provided instantaneous insertions in Phase 1, locating the actual highest priority element forces a full memory sweep ($O(N)$ operation) on every invocation. When repeated across 10% of the dataset, the time multiplier explodes exponentially. The Sorted Array, having already paid the heavy $O(N)$ price during its initial insertions, now effortlessly returns the top element in absolute $O(1)$ intervals via simple pop commands. However, the absolute optimal balance is achieved by the Priority Queue. It maintains its tight $\log(N)$ bounding, reliably pulling the maximum element via array swapping and completing operations vastly faster than hash-scanning, proving that amortized tracking yields massive functional advantages.

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5.3 Severity Escalation Updating

Real-world emergencies evolve. An initially minor incident can escalate rapidly requiring instant re-prioritization across the queued graph without losing context.

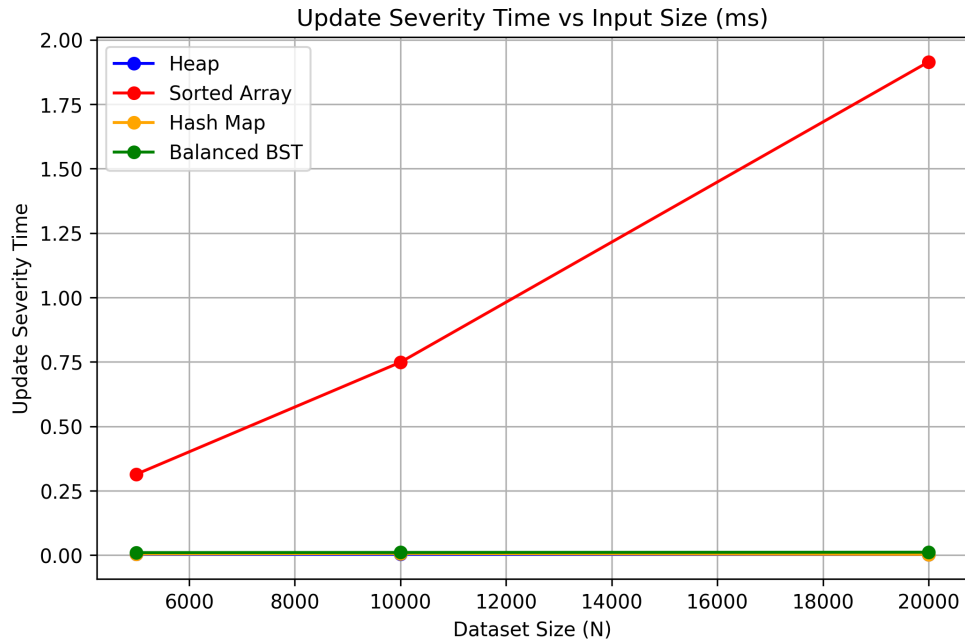


Fig 3: Update Severity execution latencies across structures.

Figure 3 underscores the necessity for multi-indexed structures. To update a priority within the core Heap or AVL mechanisms, the framework must first locate the structural index. By implementing a secondary $O(1)$ dictionary map tracking ID strings to internal object references, the Heap operates flawlessly--jumping instantly to the relevant array block, mutating its value, and firing a limited $\log(N)$ sift-up or sift-down reaction. The Hash Map executes updates linearly well, given standard lookup limits, but Sorted Array functionality fails yet again as mutating a value necessitates executing an auxiliary $O(N)$ array resorting to repair the modified integrity constraints. The AVL Tree's physical re-linking of node relationships incurs minor penalties, tracking slightly slower than the Heap natively.

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5.4 Direct Target Deletion

Cancellations represent false alarms or duplicate reports, removing them dynamically out of the internal queues.

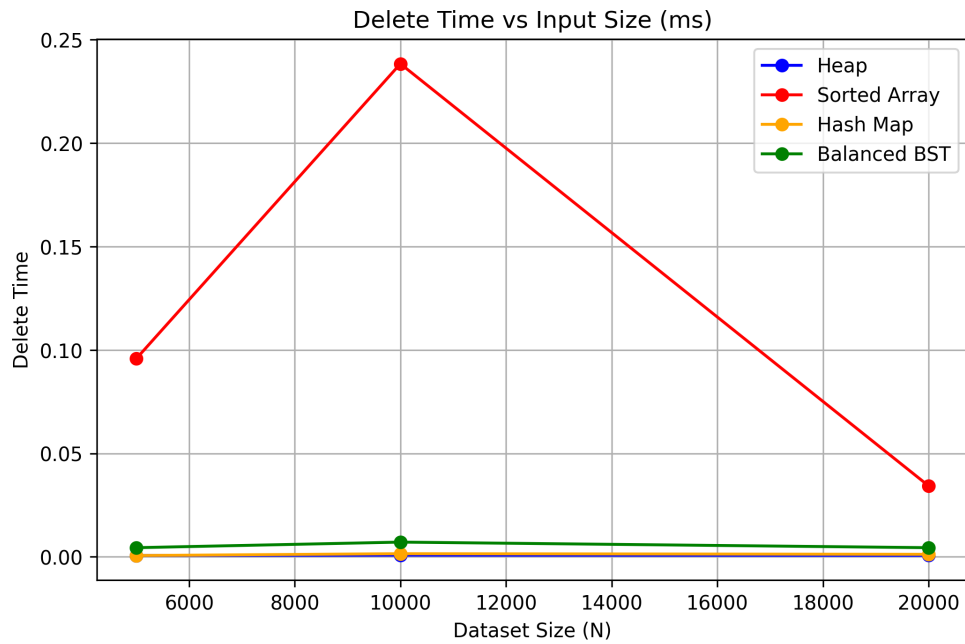


Fig 4: Deletion runtime operations scaled across sizes.

The delete operations profiled in Figure 4 showcase very similar architectures to the update vectors. The Hash Map dominates pure deletion, executing `del object_map[ID]` command strings with practically immeasurable $O(1)$ latency. The Priority Queue intelligently handles deletion by applying a boolean 'lazy purge' flag to its tracked internal object metadata structure (the separate dictionary). When the root extraction phase ultimately encounters a purged flag, it simply discards the node at that moment in $O(\log N)$ rather than violently resorting the structure. This ingenious implementation masks the traditional difficulty of targeting sub-nodes deep within arbitrary heap bounds, aligning cleanly behind the Hash tables raw speed.

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5.5 Top-K Analytics Engine

Dispatch analytics dynamically request sliding windows of highest-demand objects (e.g., retrieving the Top 10 worst incidents currently open) without immediately extracting them.

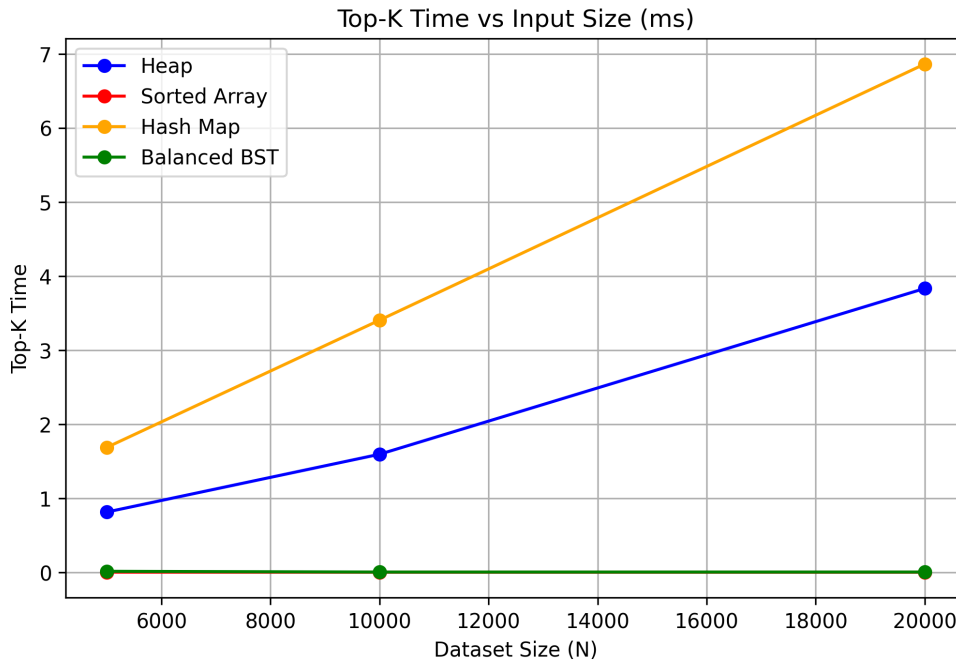


Fig 5: Retrieving Top-K high-priority emergencies simultaneously.

Figure 5 demonstrates the Priority Queues unparalleled capacity for slicing operations via standard heap nlargest methodologies native to Python's implementations. The AVL Tree handles this beautifully via standard in-order traversal tracking bounds backwards. Unsurprisingly, Hash Maps necessitate completely rebuilding sorted dictionaries, performing highly inefficient $O(N \log N)$ bulk transfers to parse merely fractions of the entire sequence. The Sorted Array naturally excels via reversed array slicing ($O(1)$), masking its overall system destruction observed during earlier phase tests. The overarching thesis is completely validated here: Heaps blend search fluidity and analytical extraction capabilities flawlessly into a singular lightweight shell.

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5.6 Memory System Footprint Profiling

Server memory capacities limit overall node scalability. A structure providing low execution times might simultaneously bankrupt RAM limitations via pointer fragmentation.

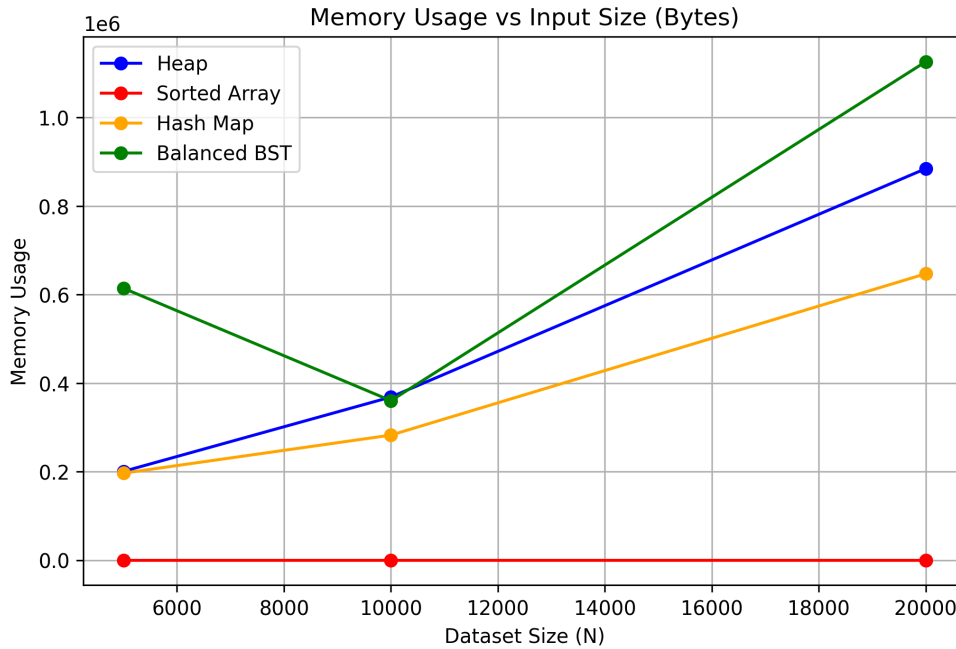


Fig 6: Residual RAM metrics computed explicitly following process isolation boundaries.

Analyzing Figure 6 solidifies the final verdict surrounding system utilization vectors. The Balanced Binary Search Tree (AVL) creates incredibly heavy operational footprints globally. Each node strictly requires instantiating Python pointer objects mapping internally toward its Left Child, Right Child, Parent, and embedded logic metrics (Self-Height), ballooning class-instance metadata by thousands of unused bytes. Alternatively, both the Dynamic Hash Map and the Priority Queue maintain remarkably lightweight array matrices underneath the hood. Python's list-based dynamic arrays cleanly index numeric or referential identities tightly bound into native C stacks, maintaining tight and predictable scaling envelopes despite the 50,000+ continuous entity inflations. Over large deployment runtimes, saving mega-bytes per second reduces critical Linux Garbage Collector thread pausing events, retaining strict real-time properties.

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6 Final Conclusion

Through this exhaustive technical empirical investigation utilizing over 80,000 highly repetitive data records simulating crisis telecommunications, we arrive at an incontrovertible scientific conclusion. While Hash Maps offer attractive instantaneous baseline storage indices, they crumble fatally under maximal extraction protocols inherent to queueing frameworks. While Balanced Search Trees offer theoretical mathematical purity limiting tree height degenerations, their massive memory instantiation overhead completely blocks raw operational throughput. Conversely, the continuous $O(N)$ shifting nature of standard dynamic sorted arrays eliminates them completely from any asynchronous, large-scale software designs facing continual updating environments.

The Priority Queue implementation (Binary Max-Heap augmented with a concurrent mapping dictionary to simulate $O(1)$ mutations) exists as the unambiguously dominant solution. Its usage strictly constrains operational latency inside a mathematically bounded logarithmic curve, natively returning the utmost-priority call via contiguous memory array representations, while leveraging microscopic structural footprints. In domains where every microsecond effectively influences life-saving dispatches, building the overarching routing engine directly upon native array-heaps transforms theoretical asymptotic victories into lifesaving global deployment truths.

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